

REPORT WEEK 1

let the sampling rate be 10000hz.

The cutoff freq should be less than half of sampling rate considering nyquist rate. So we can consider here about 2khz as cutoff freq. We need to normalise it to create the filter it is $f_c/(f_s/2)$ which is .4.

since data bit is of 8 bit lets say the first bit is sign bit thus we can represent it from -128 to +127. the window function of the fir filter here considered is rectangular. And also considered q1.7 quantization because the coefficients lies in the range of that. Here 1 is sign bit and the other 7 is fractional part. -1 to 0.9921875. 10000000 2's complement is -1.

so while desinging a filter we will multiply our 8 bit data with this coefficients which will cause it to be 16 bits and because we are having tap16 then there would be additions too. So we can make a head room of 4 bits and make the bits as 20 bits.

We have 15 order filter it have 16 taps so each coeff is 8 bits our data is 8 bits too so $8*8$ bit gives 16 bit answer. So after multipling with all taps we will be having 16 16 bit answers.

We need to add all those. So adding 16 16 bit number result in $16+\log_2(16)$ number of bits that is 20 bits.

Because if we add 2 16 bit numbers we get 17 bit number. If we add 8 16 bit number with oter 8 16 bit number we will have 8 17 bit answer similarly adding 4 17 bit with 4 17 bits get 4 18 bit like that.

So out final answer will be q6.14.

We then need to quantise it back to q1.7 format to sent it back.

I am planning to use saturation methord to the maximum for larger values in real part and for the the fractional part I am planning to add 2^6 to it and truncate it . That quantization methrod is called round to nearest quantization methord

Chosen coeff from python code is

```
[ 1.47508243e-17  4.40593836e-02  3.21811414e-02 -3.93325061e-02  
-8.18245695e-02  1.47508243e-17  1.90923995e-01  3.53992555e-01  
 3.53992555e-01  1.90923995e-01  1.47508243e-17 -8.18245695e-02  
-3.93325061e-02  3.21811414e-02  4.40593836e-02  1.47508243e-17]
```

Rectangular

taps 16

ORDER N=15

